

C6-1

1. (a) $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$

(b) $H = (X + Z) \cdot \frac{1}{\sqrt{2}}$
$$= \frac{1}{\sqrt{2}} (|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|)$$

(c) $P = |0\rangle\langle 0| + e^{i\theta} |1\rangle\langle 1|$

2. (a) 输入 $|\psi\rangle = |10\rangle$

$|1\rangle$ 经过 H 门后: $\frac{1}{\sqrt{2}} |0\rangle\langle 0| + \frac{1}{\sqrt{2}} |0\rangle\langle 1| + \frac{1}{\sqrt{2}} |1\rangle\langle 0| - \frac{1}{\sqrt{2}} |1\rangle\langle 1|$
$$= \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$$

则 control 变为 $\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

target 输入为 $|0\rangle$,

则最终输出波函数为 $\frac{1}{\sqrt{2}} |0\rangle|0\rangle - \frac{1}{\sqrt{2}} |1\rangle|1\rangle$, 即 $|\psi'\rangle = \frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |11\rangle$

(b) 输入 $|\psi\rangle = |01\rangle$

$|0\rangle$ 经过 H 门后: $\frac{1}{\sqrt{2}} |0\rangle\langle 0| + \frac{1}{\sqrt{2}} |0\rangle\langle 1| + \frac{1}{\sqrt{2}} |1\rangle\langle 0| - \frac{1}{\sqrt{2}} |1\rangle\langle 1|$
$$= \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

则 control 变为 $\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$

target 输入为 $|1\rangle$

则最终输出波函数为 $\frac{1}{\sqrt{2}} |0\rangle|1\rangle + \frac{1}{\sqrt{2}} |1\rangle|0\rangle$, 即 $|\psi'\rangle = \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle$

(c) 输入 $|11\rangle$

同 (a), $|1\rangle$ 经过 H 门后变为 $\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$

则 control 变为 $\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$

target 输入为 $|1\rangle$

则最终输出波函数为 $\frac{1}{\sqrt{2}} |0\rangle|1\rangle - \frac{1}{\sqrt{2}} |1\rangle|0\rangle$, 即 $|\psi'\rangle = \frac{1}{\sqrt{2}} |01\rangle - \frac{1}{\sqrt{2}} |10\rangle$